

~~① Matrix Game - II~~

② missing and repeating

~~③ N/3 problem~~

④ Monkeys and Doal

⑤ Ratio of boys and girls

~~Bit compression!~~

Rectangle overlap:

Christmas Tree

change character

N/3 problem.

Find N/3 majority element!

N array elements

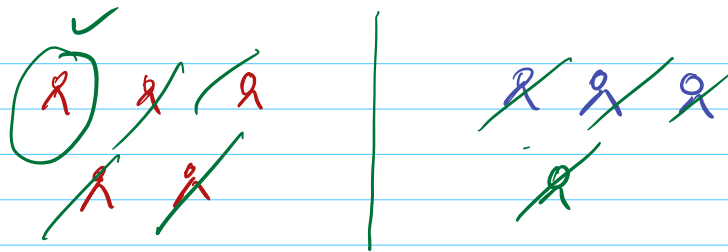
2 1 2 3 2 1 1 3

$$\left[\begin{array}{l} \text{freq}(1) = 3 > 8/3 \quad \checkmark \\ \text{freq}(2) = 3 > 8/3 \quad \checkmark \end{array} \right]$$

Ideas =

We had only one element as majority element

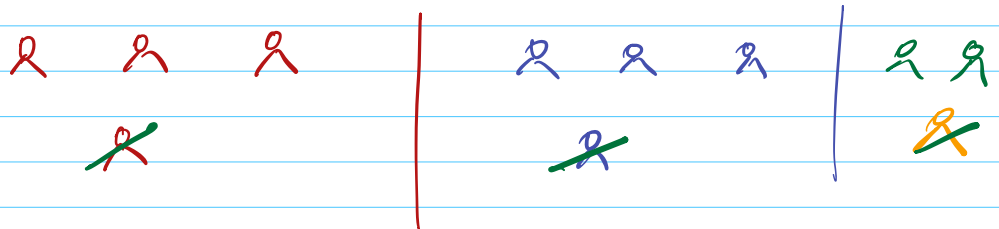
We used Moore's voting algo.
"If we remove 2 distinct elements, majority doesn't change"



$$\left| f(\text{maj}) > \frac{N}{2} \right|_{-1}^{-2}$$

$$\downarrow$$

$$f(\text{maj}) > \frac{N-2}{2}$$



$$\left| f(\text{maj}) = x > N/3 \right|$$

$$x-1 > N/3$$

$$N=26$$

$$x=9$$

$$x-1 > \frac{N}{3} - 1$$

$$9 > \frac{26}{3}$$

$$9-1 > \frac{26-2}{3}$$

$$8 > \frac{24}{3} = 8$$

Also $f > N/2 \Rightarrow$

How many majority

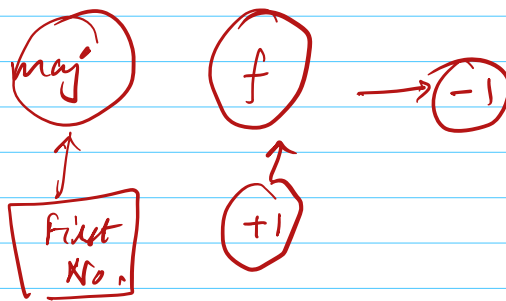
elements. $\rightarrow 2$

$$f_1 > N/3$$

$$f_2 > N/3$$

$$f_1 + f_2 > \frac{2N}{3}$$

Maj $\rightarrow -\infty$

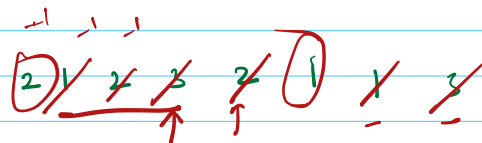


$$\text{num1} = \text{None} / -\infty$$

$$f_1 = 0$$

$$\text{num2} = \text{None} / -\infty$$

$$f_2 = 0$$



$$\text{num1} = 2$$

$$f = 2 \times 2 \times 1$$

$$\text{num2} = -\infty$$

$$f = 1 \times 2 \times 1$$

3 2 2 1 3 2 1 1
 ↑ ↑ ↑ ↑ ↑ ↑ ↑ ↑

num 1 = 3 ~~2~~ ~~2~~ ~~1~~ num 2 = 2

f = ~~0~~ ~~0~~ ~~1~~ f = ~~2~~ ~~1~~

Matrix Game 2 :

N x M

3 x 4

	1	2	3	4
1	1	2	3	4
5	5	6	7	8
6	7	10	11	12

i, j N, M
 1 1
 5 6

3 4 → 8 4

6 x 3

18 + 4

22

10⁵

① swap Col → O(N) over

② swap Row

③ $x_1 \times x_2$ $x_1, x_2 \rightarrow$ Bitwise OR
 Bitwise AND

④

my[0] → 3

my[1] → 1

2 → 1

3 → 2

i j
 1 2 3

A[0][3]

i j
 2 ↔ 3

0 ↔ 1

A[0][1] A[0][2]
 and A[1][3] → A[1][0]

A[i][0]

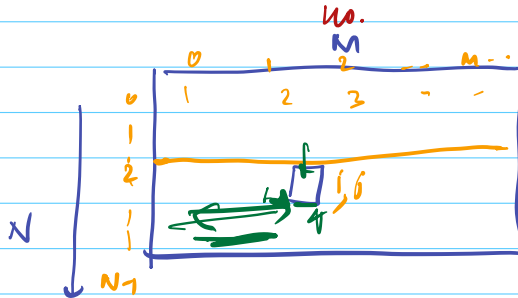
Idea 1

If given indices $i, j \rightarrow$ how to find

$N \sim 10^5$
 $M \sim 10^5$

X Matrix
can be
constructed

$N=3$
 $M=4$

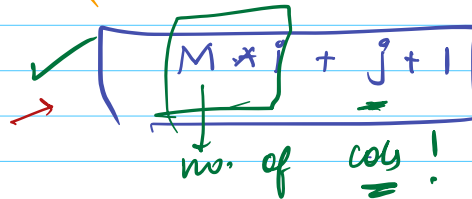


$(i) \times M$

	0	1	2	3
0	0	1	2	3
1	4	5	6	7
2	8	9	10	11

$(2, 1)$

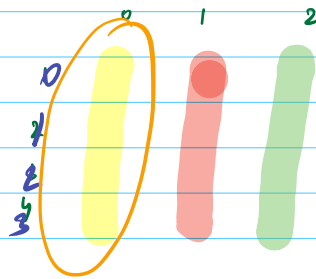
$$2 \times 4 + 2 = 10$$



Idea 2

$\rightarrow$ If there would have been only column swaps $\rightarrow$ can we find what value is present at (i, j)

$0 \leftrightarrow 1$	1	0
$1 \rightarrow 1$	0	2
$2 \rightarrow 2$	2	0



1	2	3
4	5	6
7	8	9
10	11	12

0	1
2	1
2	0

$0 \leftrightarrow 1$

0 1 2

arr

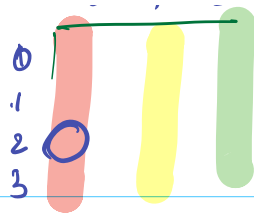
$0 \rightarrow 1$
 $1 \rightarrow 2$
 $2 \rightarrow 0$

0

$0 \rightarrow 0$

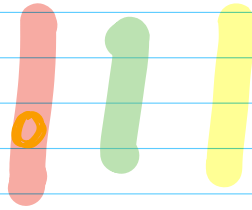
$1 \rightarrow 2$

$2 \rightarrow 1$



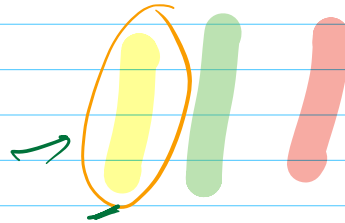
$\begin{matrix} 2 & 1 & 3 \\ 5 & 7 & 6 \\ 11 & 10 & 1 \end{matrix}$
 $\times \textcircled{6} \neq \begin{bmatrix} 2 \\ 2 \end{bmatrix} \begin{bmatrix} 0 \end{bmatrix}$

4×3



$\begin{bmatrix} 2 \end{bmatrix} \begin{bmatrix} 1 \end{bmatrix} \rightarrow 3 \times 2 + 1 + 1$

$\begin{bmatrix} 2 \end{bmatrix} \begin{bmatrix} 0 \end{bmatrix} \rightarrow 2 \textcircled{4}$



✓ $\text{rel_mapping} = \{3\}$

$0 \rightarrow 0$
 $1 \rightarrow 1$
 $2 \rightarrow 2$
 $\vdots$
 $M-1 \rightarrow M-1$

only rows

↓

row_swapping

would work.

1 ↔ 2

2 ↔ 3

0 ↔ 2 col

1 ↔ 3 Row

4x3

	0	1	2
0	1	2	3
1	4	5	6
2	7	8	9
3	10	11	12

0 → 1
1 → 2
2 → 0

0 → 0
1 → 2
2 → 3
3 → 1

↓ swap col → 1, 2

1	3	2	
4	6	5	
7	9	8	← → 10 12
10	12	11	← → 7 9 8

[2] [2]

↓
3
↓
1

3x3 + 1 + 1

11

2 3 1

5 6 4

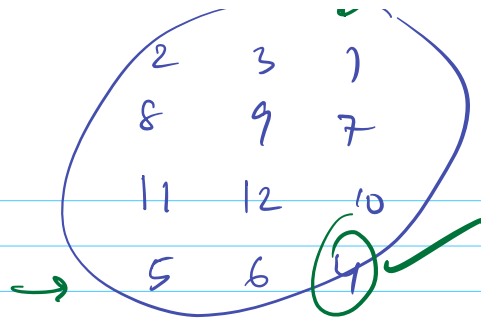
11 12 10

8 9 7

3, 2
↓
1 0

1x3 + 0 + 1

= 4



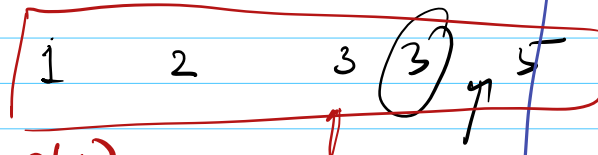
$$M \times i + j + 1$$

i, j

$$M \times \text{row}[i] + \text{col}[j] + 1$$

Missing and Repeating Number:

1 $\rightarrow$ N



(7) $O(N)$
(8) $O(1)$

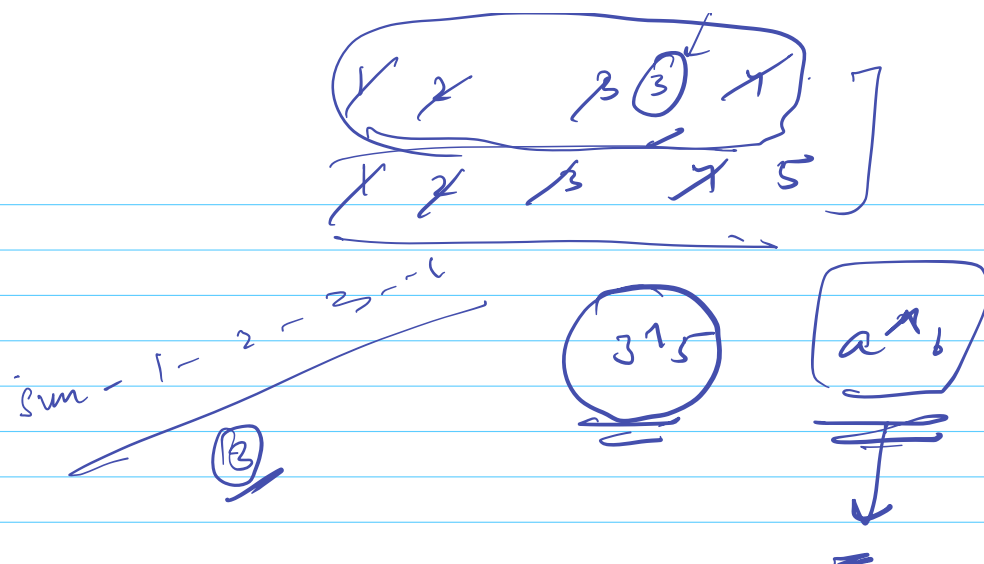
(N+1) numbs
1 $\rightarrow$ N
repeating

sort
 $\rightarrow n \log n$

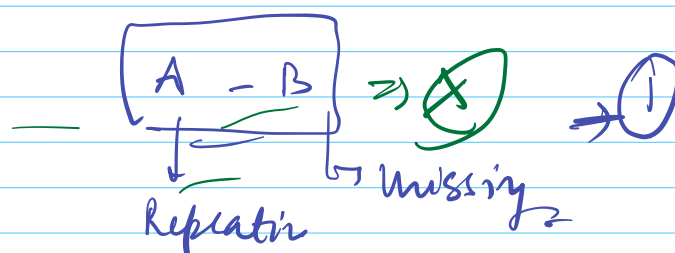
Sum of
N
num

Frequency Array

$O(N)$
 $O(N) \rightarrow O(1)$



✓ Sum of all array elements
 - $\left(\text{Sum of } n \text{ natural} \right)$
 $\frac{n \times (n+1)}{2}$



sum of squares of all array element

- $\left(\text{sum of square of } n \text{ natural} \right)$

✓ $\boxed{A^2 - B^2} \Rightarrow \checkmark$

$(A-B)(A+B) = \textcircled{Y}$

$\frac{n \times (n+1) \times (2n+1)}{6}$

$$X (A+B) = Y$$

$$A+B = \frac{Y}{X}$$

Step 1

$$1 \quad 2 \quad 3 \quad 3 \quad 5 \rightarrow X \quad - (1)$$

$$1 \quad 2 \quad 3 \quad 4 \quad 5 \rightarrow Y \quad - (2)$$

$$(1) - (2)$$

$$R - M = V_1$$

$$- (3)$$

$$1^2 \quad 2^2 \quad 3^2 \quad 3^2 \quad 5^2 =$$

$$1^2 \quad 2^2 \quad 3^2 \quad 4^2 \quad 5^2$$

$$3^2 - 4^2$$

$$R^2 - M^2 = V_2$$



$$(R-M) \times (R+M) = V_2$$

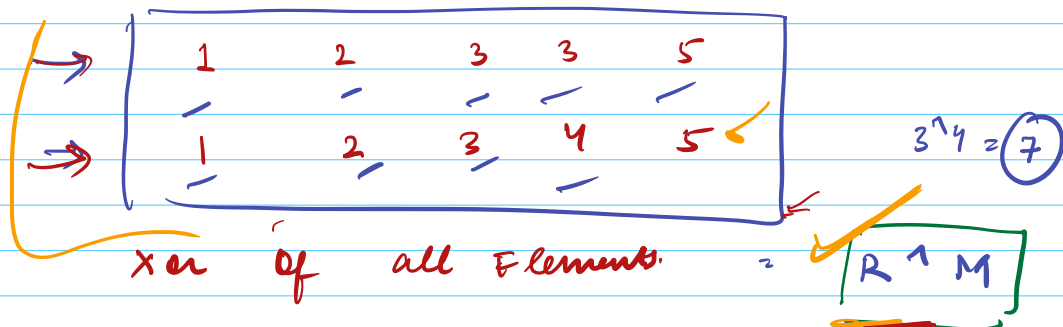
using Eq (3)

$$V_1 \times (R+M) = V_2$$

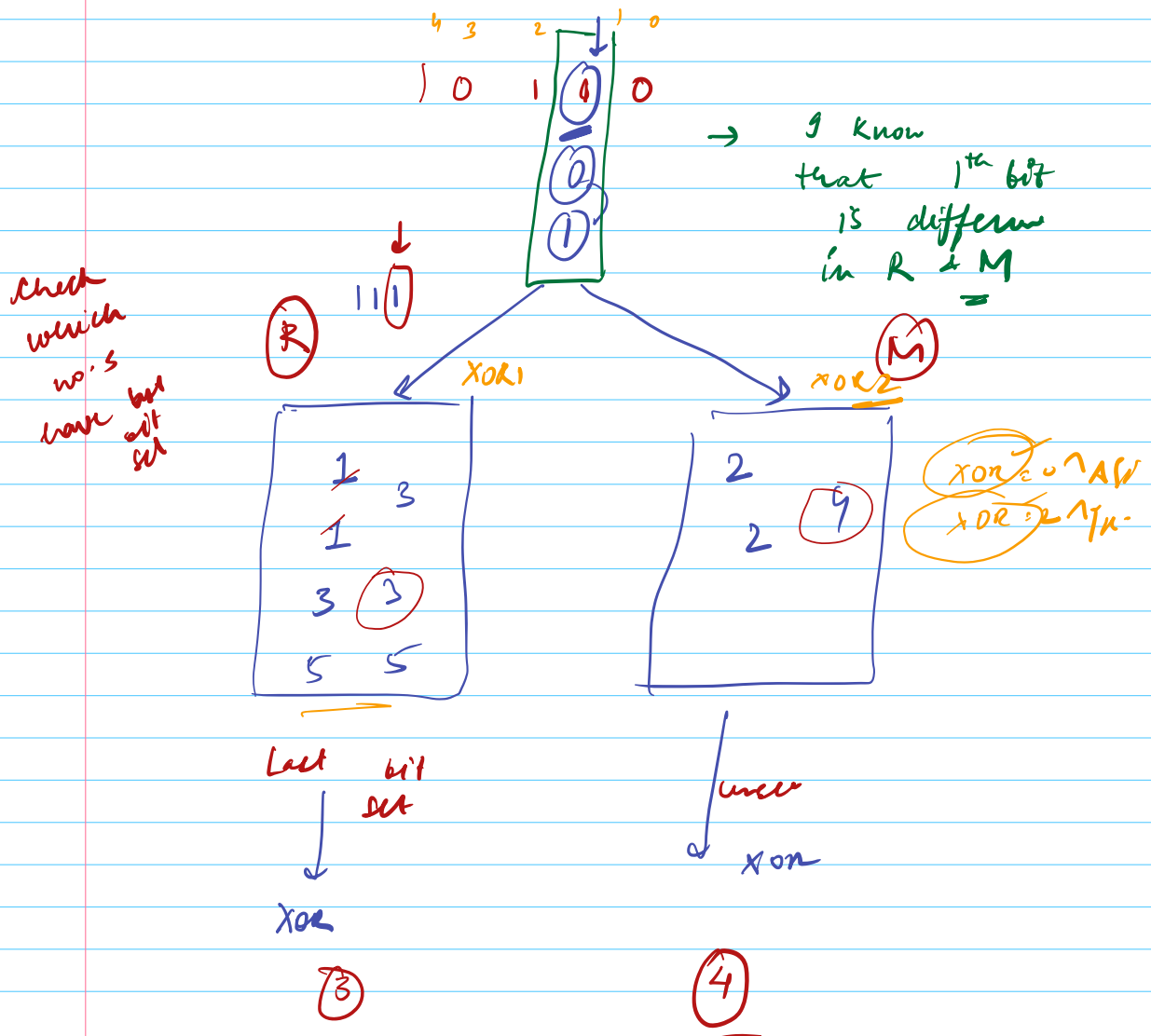
$$R+M = \frac{V_2}{V_1} \quad - (*)$$

using Eq (*), (3)

$$R-M = \frac{R+M}{R \cdot M}$$



(4) If a bit is '1' in xor.
 It means the bit is 1 in one number and 0 in other.



$\begin{matrix} & & & & & & & a & a & & \\ & & & & & & \nearrow & \nearrow & \nearrow & & \\ a & a & a & a & b & b & b & c & c & d & \\ & & & & & & & & & & \Rightarrow 2 \end{matrix}$

3 charact

